

Ahmed M. Ali

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PERSONAL SUMMARY

Senior Software Engineer with several-years experience developing perception and autonomy systems for UAVs operating in **GPS-denied environments**. Strong background in deep learning (PyTorch), reinforcement learning, and real-time C++/Python systems deployed on **embedded edge platforms**. My expertise lies in translating complex research results into robust, **production-ready software solutions**. I also hold an Austrian **Red-White-Red+ card** (unrestricted work authorization) and a valid **Austrian driving license**.

WORK EXPERIENCE

Senior Software Engineer - Team Lead – *Avemoy, Austria* Sep 2024 – Jan 2026

- Developed and deployed **perception-driven UAV autonomy modules**, processing LiDAR, point cloud, and visual data for real-time navigation in GPS-denied environments.
- Implemented and integrated **multi-sensor fusion** between **point cloud, LiDAR, and VIO**, reducing altitude estimation error by 20% in real flight conditions.
- Optimized perception and state-estimation pipelines by migrating performance-critical components from **Python to C++**, reducing CPU load by **50%** on embedded edge platforms.

Research Engineer – *University of Klagenfurt, Austria* Jul 2023 – Jul 2024

- Developed a **point-cloud-based** altitude estimation module, increasing long autonomous mission success rate for drones by 60% in GPS-denied environments.
- Optimized autonomy pipeline by converting critical modules from Python to C++, reducing onboard CPU load by 50% on edge hardware
- Built trajectory **evaluation tools** to estimate flight accuracy, supporting OptiTrack and real-time flight data.

SKILLS

- Programming:** Python, C++, Bash, Matlab
- AI / ML:** PyTorch, TensorFlow, Reinforcement Learning, CNNs, YOLO
- Computer Vision:** OpenCV, YOLO, Image Processing, Feature Extraction
- Robotics:** ROS/ROS2, MAVROS, ArduPilot, SLAM, Motion Planning
- Tools:** Docker, Git, CI/CD, Jira
- Embedded Systems:** Raspberry Pi, Odroid, NVIDIA Jetson
- Leadership:** Project Management, Client Communication, Agile Development
- Languages:** Arabic (Native), English(C1), German (B1)

EDUCATION

Master of Science in Robotics and Computer Vision Aug 2021 – Jun 2023

- Innopolis University, Russia.*
- Accumulative GPA: **4.94 / 5.00**. **"High Honors"**.
- Master thesis: "Trajectory Planner for Autonomous Ground Vehicles in Unknown Environments". [Report](#)

- *Nile University, Egypt.*
- Accumulative GPA: **3.97 / 4.00**. "**High Honors**".
- Bachelor thesis: "Designing and Control Optimization of Autonomous Mobile Industrial Robots".
[Report](#)

SELECTED PROJECTS

Autonomous Greenhouse Inspection Drone, Team Lead

- Led navigation team in developing **GPS-free autonomous** flight system adapted from NASA's Ingenuity Mars helicopter algorithm, enabling real-time 4k field scanning of greenhouse crops in **semi-unknown environments**
- Optimization of perception and control algorithms to minimize computational effort while meeting real-time requirements.
- Delivered **production-ready MVP** using ROS2, Docker, C++, and Python, implementing autonomous decision-making systems for continuous scanning and charging cycle

Human Iris Center Detection – Deep Learning (PyTorch)

[Repo](#)

- Implemented CNN model based on published research for iris localization.
- Applied morphological preprocessing and Gaussian filtering to improve robustness.
- Built complete training and evaluation pipeline in PyTorch.

Robot Navigation Using Reinforcement Learning

[Repo](#)

- Implemented continuous control navigation using DDPG and TD3 algorithms.
- Built custom 2D simulation environment using OpenCV and OpenAI Gym.
- Trained and evaluated RL models for stable trajectory generation.

PUBLICATIONS

- **Ali** et al., Heliyon, 2023 – Systematic review of AGVs with mounted manipulators. ([Paper](#))
- **Ali** et al., IGI Global, 2022 – Battery degradation in EVs. ([Paper](#))